

CARLSBAD CAVERN CITY, NM, USA

WMO: 722687

Lat: 32.334N

Lon: 104.258W

Elev: 985

StdP: 90.04

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-6.3	-4.3	-18.6	0.8	8.7	-16.5	1.0	9.6	15.1	13.2	13.0	13.7	2.5	310	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	39.6	18.3	38.0	18.5	36.8	18.5	22.0	30.1	21.5	29.9	21.0	29.7	5.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.3	16.9	23.8	19.5	16.1	23.6	18.8	15.4	23.5	69.4	29.7	67.1	29.8	65.2	29.6	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	10.9	9.1	-10.4	42.3	2.8	1.2	-12.4	43.2	-14.1	43.9	-15.7	44.6	-17.7	45.5		
(5)			-11.4	23.3	2.5	1.1	-13.3	24.1	-14.7	24.7	-16.2	25.3	-18.0	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	7.1	9.6	13.6	18.1	23.1	27.7	28.3	27.6	23.7	17.9	11.4	6.8		
	DBStd	8.75	4.73	5.10	4.74	4.33	3.87	2.58	2.29	2.29	3.18	4.13	4.49	4.71		
	HDD10.0	359	113	62	22	3	0	0	0	0	3	37	120			
	HDD18.3	1442	347	245	157	56	9	0	0	0	3	58	210	357		
	CDD10.0	3261	24	51	132	246	406	531	568	545	412	247	79	22		
	CDD18.3	1302	0	1	9	48	157	281	309	287	165	43	2	0		
	CDH23.3	17163	10	64	298	886	2172	3834	3965	3469	1766	606	86	8		
	CDH26.7	9329	1	11	82	363	1175	2337	2312	1947	865	224	13	0		
(14) Wind	WSAvg	4.0	3.9	4.3	4.6	5.0	4.6	4.3	3.6	3.3	3.4	3.6	3.7	3.8		
(15) Precipitation	PrecAvg	330	11	13	10	15	29	37	44	50	60	31	19	14		
	PrecMax	615	44	57	55	106	107	144	190	196	312	147	116	96		
	PrecMin	113	0	0	0	0	0	0	1	1	0	0	0	0		
	PrecStd	128	10	14	12	23	27	35	43	47	67	35	29	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	28.3	31.2	34.7	38.9	42.2	40.9	39.9	37.6	33.6	28.4	24.5		
		MCWB	10.3	11.7	12.6	14.3	17.2	18.7	19.1	19.5	18.6	16.2	12.9	10.3		
	2%	DB	22.3	25.3	29.0	32.4	37.1	40.2	38.8	37.8	35.3	31.4	25.9	22.0		
		MCWB	9.3	10.4	11.9	13.3	16.4	17.7	19.3	19.4	18.1	16.0	12.5	9.5		
	5%	DB	19.8	22.8	27.0	30.7	35.1	38.6	37.4	36.4	33.8	29.2	23.6	19.1		
		MCWB	8.5	9.5	11.2	12.9	15.5	17.7	19.4	19.4	18.3	15.6	11.2	8.5		
	10%	DB	17.2	20.2	24.3	28.8	33.0	36.7	35.9	35.0	31.8	27.0	21.1	16.5		
		MCWB	7.1	8.5	10.3	12.5	15.1	17.9	19.3	19.7	18.3	15.0	10.6	7.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	13.1	15.5	17.6	20.3	22.2	23.1	22.8	22.3	21.2	15.8	12.1		
		MCDB	20.3	22.8	21.9	25.1	30.6	32.7	29.6	30.2	27.2	25.7	21.8	20.3		
	2%	WB	10.2	11.7	13.7	16.0	19.2	21.3	21.9	21.9	21.2	19.4	13.9	10.4		
		MCDB	19.6	22.5	23.6	24.6	29.1	31.6	30.5	29.8	28.0	25.2	21.9	19.3		
	5%	WB	8.9	10.3	12.4	14.9	18.4	20.6	21.4	21.4	20.5	18.1	12.6	9.1		
		MCDB	18.7	20.9	23.2	24.8	28.4	30.9	30.2	29.7	27.8	23.9	21.0	17.9		
	10%	WB	7.6	9.1	11.2	13.6	17.5	20.0	20.8	20.8	19.9	16.7	11.3	7.8		
		MCDB	16.3	19.3	22.2	25.4	27.4	30.5	29.9	29.6	27.1	23.4	19.6	15.9		
(35) Mean Daily Temperature Range	MDBR		15.7	16.1	16.8	17.2	16.5	16.4	14.1	13.9	14.0	15.3	15.9	14.8		
	5% DB	MCDBR	20.3	21.0	21.3	20.5	19.8	19.8	16.4	16.0	17.3	19.2	20.3	19.7		
		MCWBR	10.7	10.2	9.3	8.7	8.1	6.4	4.6	4.5	5.5	7.8	9.9	10.2		
	5% WB	MCDBR	18.8	19.2	18.7	17.7	16.8	16.0	13.6	13.4	13.2	14.1	17.3	17.7		
		MCWBR	10.4	9.6	8.7	8.2	7.3	5.4	3.9	3.9	4.5	6.2	8.9	9.5		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.262	0.280	0.317	0.331	0.357	0.383	0.364	0.341	0.295	0.273	0.259		
	taud		2.616	2.591	2.530	2.401	2.428	2.404	2.377	2.423	2.441	2.578	2.566	2.591		
	Ebn at Noon		992	1010	1005	971	954	923	898	913	925	957	955	965		
	Edh at Noon		76	86	99	118	116	119	122	114	108	87	79	74		
(44) All-Sky Solar Radiation	RadAvg		3.56	4.48	5.77	7.05	7.58	7.74	7.22	6.64	5.74	4.72	3.73	3.15		
	RadStd		0.31	0.38	0.35	0.41	0.43	0.34	0.34	0.29	0.37	0.42	0.33	0.24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.73	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page